Born-Oppenheimer Approximation

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The Born-Oppenheimer Hamiltonian. The Born-Oppenheimer (BO) Hamiltonian omits the kinetic energy of the nuclei. Here, i, j label the N electrons with positions r_i, and I, J label the M nuclei with positions R_I and charges $Z_I e$. Both r_i and R_I are vectors of any dimension, and $||$ is the Euclidean norm. The Hamiltonian is

$$\boxed{\begin{aligned}\hat{\mathcal{H}}^{\mathrm{BO}} = & \underbrace{-\sum_{i=1}^{N} \frac{\hbar^2}{2m_e} \nabla_{r_i}^2}_{\text{electron kinetic energy}} + \underbrace{\frac{1}{2} \sum_{I \neq J}^{M} \frac{e^2 Z_I Z_J}{4\pi\epsilon_0 \left|R_I - R_J\right|}}_{\text{nuclei-nuclei repulsion}} \\ & \underbrace{- \sum_{i=1}^{N} \sum_{I=1}^{M} \frac{e^2 Z_I}{4\pi\epsilon_0 \left|R_I - r_i\right|}}_{\text{nuclei-electron attraction}} + \underbrace{\frac{1}{2} \sum_{i \neq j}^{N} \frac{e^2}{4\pi\epsilon_0 \left|r_i - r_j\right|}}_{\text{electron-electron repulsion}} .\end{aligned}} \tag{0.1}$$

The full Hamiltonian. Restoring the kinetic energy of the nuclei gives the full Hamiltonian:

$$\boxed{\hat{\mathcal{H}} = -\sum_{I=1}^{M} \frac{\hbar^2}{2M_I} \nabla_{R_I}^2 + \hat{\mathcal{H}}^{\mathrm{BO}} = \hat{P}_R \, \hat{\mathcal{M}}^{-1} \hat{P}_R + \hat{\mathcal{H}}^{\mathrm{BO}}.} \tag{0.2}$$

Here, $R = (R_1, \ldots, R_M)$ collects all nuclear coordinates, and $\hat{\mathcal{M}}$ denotes the mass matrix, to distinguish it from the number of nuclei M. The gradient ∇_R collects the derivatives with respect to every nuclear coordinate. In d dimensions, it has dM components, which is $3M$ in three dimensions. The inverse mass matrix $\hat{\mathcal{M}}^{-1}$ is diagonal and assigns the factor $1/M_I$ to each of the d components of nucleus I. Therefore, the inverse mass operator acts on the momentum operator and not the wavefunction. The momentum operator can also act on the inverse mass operator by simply collecting the corresponding mass term. So the momentum operator and inverse mass operator commute:

$$\boxed{[\hat{P}, \hat{\mathcal{M}}^{-1}] = 0.} \tag{0.3}$$

The Born-Oppenheimer equation and surfaces. Within the BO Hamiltonian, the nuclear coordinates R are fixed parameters. For each fixed R, the electrons obey the Born-Oppenheimer equation

$$\boxed{\hat{\mathcal{H}}_{\mathrm{BO}} \left|\psi_n^{\mathrm{BO}}(R)\right\rangle = E_n^{\mathrm{BO}}(R) \left|\psi_n^{\mathrm{BO}}(R)\right\rangle.} \tag{0.4}$$

The eigenstate depends on the electron coordinates r and, parametrically, on R. The ket $|\psi_n^{\mathrm{BO}}(R)\rangle$ stands for the full function $\psi_n^{\mathrm{BO}}(r, R)$, where $r = (r_1, \ldots, r_N)$ collects all electron coordinates. Varying R traces out the eigenenergy E_n^{BO} as a function of the dM nuclear coordinates. This function is called the n-th Born-Oppenheimer surface, or potential energy surface. It includes the nuclei-nuclei repulsion, and it acts as the potential for nuclear motion in the next step.

The Born-Huang expansion. Let $\Psi(r, R, t)$ be a solution of the time-dependent Schrödinger equation of the full Hamiltonian. For each fixed R, the BO eigenstates form a complete basis for the electrons, so we can expand

$$\boxed{|\Psi(r, R, t)\rangle = \sum_n |\psi_n^{\mathrm{BO}}(R)\rangle\, \Theta_n(R, t).} \tag{0.5}$$

The coefficient $\Theta_n(R, t)$ is the nuclear wavefunction associated with the n-th electronic state, that is, with the n-th BO surface. It does not depend on r, because it is the coefficient of the r-dependent states. Its square modulus, $|\Theta_n(R, t)|^2$, is the probability density of finding the nuclei at R at time t with the electrons in the n-th BO state. This expansion is called the Born-Huang expansion.

Time evolution in the BO basis. To study the dynamics, we insert the Born-Huang expansion into the time-dependent Schrödinger equation:

$$i\hbar\, \partial_t |\Psi(r, R, t)\rangle = \Big(\hat{\mathcal{H}}_{\mathrm{BO}} + \tfrac{1}{2}\, \hat{P}\hat{\mathcal{M}}^{-1}\hat{P}\Big)|\Psi(r, R, t)\rangle. \tag{0.6}$$

On the left side, the BO eigenstates are stationary, so ∂_t acts only on the nuclear wavefunctions:

$$i\hbar\, \partial_t \sum_n |\psi_n^{\mathrm{BO}}(R)\rangle\, \Theta_n(R, t) = \sum_n |\psi_n^{\mathrm{BO}}(R)\rangle\; i\hbar\, \partial_t \Theta_n(R, t). \tag{0.7}$$

To isolate the dynamics of one nuclear wavefunction, we project onto $\langle\psi_m^{\mathrm{BO}}|$. The BO eigenstates are orthonormal at every R, $\langle\psi_m^{\mathrm{BO}}|\psi_n^{\mathrm{BO}}\rangle = \delta_{mn}$, so

$$\langle\psi_m^{\mathrm{BO}}|\, i\hbar\, \partial_t |\Psi\rangle = i\hbar\, \partial_t \Theta_m(R, t). \tag{0.8}$$

The first term of the right side is diagonal in the BO basis:

$$\hat{\mathcal{H}}_{\mathrm{BO}} \sum_n |\psi_n^{\mathrm{BO}}(R)\rangle\, \Theta_n(R, t) = \sum_n E_n^{\mathrm{BO}}(R)\, |\psi_n^{\mathrm{BO}}(R)\rangle\, \Theta_n(R, t), \tag{0.9}$$

and its projection gives $E_m^{\mathrm{BO}}(R)\, \Theta_m(R, t)$. The second term needs care. An operator acts on the rightmost thing to it, and here both $|\psi_n^{\mathrm{BO}}(R)\rangle$ and $\Theta_n(R, t)$ depend on R. Hence $\hat{P}$ obeys the product rule, and applying it twice gives

$$\hat{P}\hat{\mathcal{M}}^{-1}\hat{P}\big(|\psi_n^{\mathrm{BO}}\rangle\, \Theta_n\big) = \hat{P}\hat{\mathcal{M}}^{-1}\hat{P}|\psi_n^{\mathrm{BO}}\rangle\Theta_n + 2\hat{P}|\psi_n^{\mathrm{BO}}\rangle\hat{\mathcal{M}}^{-1}\hat{P}\Theta_n + |\psi_n^{\mathrm{BO}}\rangle\, \hat{P}\hat{\mathcal{M}}^{-1}\hat{P}\Theta_n. \tag{0.10}$$

Projecting onto $\langle\psi_m^{\mathrm{BO}}|$ and using orthonormality on the last term gives

$$\frac{1}{2}\langle\psi_m^{\mathrm{BO}}|\hat{P}\hat{\mathcal{M}}^{-1}\hat{P}|\Psi\rangle = \frac{1}{2}\hat{P}\hat{\mathcal{M}}^{-1}\hat{P}\,\Theta_m + \sum_n \left(\langle\psi_m^{\mathrm{BO}}|\hat{P}|\psi_n^{\mathrm{BO}}\rangle\hat{\mathcal{M}}^{-1}\hat{P}\Theta_n + \frac{1}{2}\langle\psi_m^{\mathrm{BO}}|\hat{P}\hat{\mathcal{M}}^{-1}\hat{P}|\psi_n^{\mathrm{BO}}\rangle\,\Theta_n\right). \tag{0.11}$$

Therefore, the full nuclear Schrödinger equation is

$$\boxed{\begin{aligned} i\hbar\,\partial_t\Theta_m(R,t) = &\left(E_m^{\mathrm{BO}}(R) + \frac{1}{2}\hat{P}\hat{\mathcal{M}}^{-1}\hat{P}\right)\Theta_m(R,t) \\ &+ \sum_n \left[\langle\psi_m^{\mathrm{BO}}|\hat{P}|\psi_n^{\mathrm{BO}}\rangle\hat{\mathcal{M}}^{-1}\hat{P}\Theta_n + \frac{1}{2}\langle\psi_m^{\mathrm{BO}}|\hat{P}\hat{\mathcal{M}}^{-1}\hat{P}|\psi_n^{\mathrm{BO}}\rangle\,\Theta_n(R,t)\right]. \end{aligned}} \tag{0.12}$$

Non-adiabatic coupling. We can see that the evolution of the m-th excited state involves other states. This is because R is a function of t. As time evolves, the BO eigenbasis also change. A BO eigenbasis at time $t+\delta t$ can be expressed as a linear combination of BO eigenbasis at time t. That's why time evolution requires coupling between eigenbasis. The first term is called the first derivative coupling, denoted as

$$\boxed{\vec{A}_{mn} = \langle\psi_m^{\mathrm{BO}}|\hat{P}|\psi_n^{\mathrm{BO}}\rangle.} \tag{0.13}$$

It's a dM dimensional vector field because the momentum operator is a gradient. It is hermitian

$$\begin{aligned} \hat{P}\langle\psi_m^{\mathrm{BO}}|\psi_n^{\mathrm{BO}}\rangle &= \langle\psi_m^{\mathrm{BO}}|\hat{P}\psi_n^{\mathrm{BO}}\rangle - \langle\hat{P}\psi_m^{\mathrm{BO}}|\psi_n^{\mathrm{BO}}\rangle \\ &= \vec{A}_{mn} - \vec{A}_{nm}^{*} \\ &= 0 \end{aligned} \tag{0.14}$$

The second term is called the second derivative coupling, denoted as

$$\boxed{B_{mn} = \frac{1}{2}\langle\psi_m^{\mathrm{BO}}|\hat{P}\hat{\mathcal{M}}^{-1}\hat{P}|\psi_n^{\mathrm{BO}}\rangle.} \tag{0.15}$$

We can rewrite the second derivative coupling using the first derivative coupling. Applying the product rule to the vector field $\vec{A}_{mn}$ gives

$$\hat{\mathcal{M}}^{-1}\hat{P}\vec{A}_{mn} = \langle\psi_m^{\mathrm{BO}}|\hat{P}\hat{\mathcal{M}}^{-1}\hat{P}|\psi_n^{\mathrm{BO}}\rangle - \langle\hat{\mathcal{M}}^{-1}\hat{P}\psi_m^{\mathrm{BO}}|\hat{P}\psi_n^{\mathrm{BO}}\rangle. \tag{0.16}$$

The sign is negative because the bra of $\hat{P}\psi_m^{\mathrm{BO}}$ carries the complex conjugate of $-i\hbar$. Inserting the completeness relation between the two vectors

$$\hat{\mathcal{M}}^{-1}\hat{P}\vec{A}_{mn} = \langle\psi_m^{\mathrm{BO}}|\hat{P}\hat{\mathcal{M}}^{-1}\hat{P}|\psi_n^{\mathrm{BO}}\rangle - \sum_j \langle\hat{\mathcal{M}}^{-1}\hat{P}\psi_m^{\mathrm{BO}}|\psi_j^{\mathrm{BO}}\rangle\langle\psi_j^{\mathrm{BO}}|\hat{P}\psi_n^{\mathrm{BO}}\rangle. \tag{0.17}$$

Using the hermiticity of the first derivative coupling, we obtain the final form

$$\boxed{B_{mn} = \frac{1}{2}\,\hat{\mathcal{M}}^{-1}\hat{P}\vec{A}_{mn} + \frac{1}{2}\sum_j \mathcal{M}^{-1}\vec{A}_{jm}^{*}\vec{A}_{jn}.} \tag{0.18}$$

The full Hamiltonian rewritten. The full Hamiltonian has the final form

$$\boxed{\hat{\mathcal{H}} = \hat{\mathcal{H}}^{\mathrm{BO}} + \frac{1}{2}\mathcal{M}^{-1}(\hat{P} + \hat{A})^2.} \tag{0.19}$$